The complete exclusion list for simple string C-groups

Human Compression project: recovery of L011

We recover the principal classification of Leemans and Vandenschrick [1]. The complete companion *Projective involutions and support spaces in small dimension* is an acquired common proof. The finite exclusions below are reproducible exhaustive computations, with a proved enumeration boundary. The positive direction retains the deep generation classification attributed by the source to Nuzhin and Mazurov; it is not reproved by these computations.

1 The retained positive classification and the intersection property

We retain the following exact statement of the established rank-three generation results, as quoted in the introduction of [1]: every nonabelian finite simple group outside

$$\begin{gathered}\mathrm{PSL}_3(q),\quad \mathrm{PSU}_3(q),\quad \mathrm{PSL}_4(2^n),\quad \mathrm{PSU}_4(2^n),\\ \mathrm{PSU}_4(3),\quad \mathrm{PSU}_5(2),\quad A_6,\quad A_7,\quad M_{11},\quad M_{22},\quad M_{23},\quad \mathrm{McL}\end{gathered} \tag{1}$$

is generated by three involutions with two commuting. Here q ranges over prime powers and $n \geq 1$, and only the nonabelian simple members of the listed families are relevant. The source attributes the classical and alternating cases to Nuzhin, the sporadic cases to Mazurov, and the additional exceptions $\mathrm{PSU}_4(3), \mathrm{PSU}_5(2)$ to Jones and Mačaj (personal communication). Our finite checks independently establish their negative direction. The positive classification remains borrowed, including its classification-theoretic hypotheses and original proof costs. Names in (1) specify abstract isomorphism types; coincidences of different classical names do not create new cases.

The following elementary proof supplies the rank-three intersection step in the simple case. The source cites the more general Conder–Oliveros theorem.

Lemma 1. *Let a nonabelian finite simple group G be generated by involutions a, b, c with a, c commuting. Then (a, b, c) is a rank-three string C-group marking.*

Proof. The generators are distinct and neither adjacent pair commutes: otherwise the generated group would be dihedral or would have a nontrivial commuting normal factor. Put $A = \langle a, b\rangle$, $B = \langle b, c\rangle$ and $H = A \cap B$. Both A, B are proper dihedral subgroups. If $H > \langle b\rangle$, then H is a dihedral group, since a subgroup of a dihedral group containing its reflection b consists of a cyclic rotation subgroup and its coset by b. Let $X = H \cap \langle ab\rangle$ and $Y = H \cap \langle bc\rangle$. If H is not a four-group, its cyclic index-two rotation subgroup is unique, so $X = Y \neq 1$. This subgroup is normal in both A and B, since every subgroup of a dihedral rotation subgroup is normal. It is then a nontrivial cyclic normal subgroup of G, a contradiction.

If H is a four-group and $X = Y$, the same argument applies. Otherwise write $X = \langle u\rangle$, $Y = \langle v\rangle$; its third involution is $b = uv$. The element u is central in A and v is central in B. Thus every one of a, v commutes with every one of c, u. It follows that $G = \langle a, v\rangle\langle c, u\rangle$ is a product of two commuting nontrivial normal subgroups, impossible for a nonabelian simple group. Consequently $A \cap B = \langle b\rangle$.

The endpoint subgroup is $\langle a, c\rangle \cong C_2^2$. Its intersection with A contains a; if larger, it contains c and would give $G = A$. The dual assertion holds for B. These three intersections, and independence (which follows since no two involutions generate G), give all subset intersections in rank three. □

2 A complete finite elimination method

For a fixed faithful finite permutation group G, we enumerate ordered prefixes $p = (t_0, \ldots, t_{r-1})$ of connected string C-group markings, up to simultaneous conjugation by G. A prefix is retained only if it has the full intersection property. Define

$$H(p) = \langle t_0, \ldots, t_{r-1} \rangle, \quad Z(p) = C_G(t_0, \ldots, t_{r-1}), \quad E(p) = \{x \in C_G(t_0, \ldots, t_{r-2}) : |x| = 2\}.$$

For length one the last set means all involutions of G. The algorithm starts with one representative of each involution conjugacy class. At p it filters $E(p)$ by $x \notin H(p)$ and $|t_{r-1}x| > 2$, then takes one representative of every $Z(p)$-orbit under conjugation. For each representative it tests independence and all subset intersections of the extended prefix. A cheap adjacent-dihedral intersection check is applied first; it is only an early necessary rejection. If the extension generates G it is recorded. Otherwise we recurse. The next eligible set is computed by filtering $E(p)$ for commutation with t_{r-1}, so is the entire involution set of $Z(p)$.

Proposition 2. *This algorithm terminates and finds every connected string C-group marking of G, up to simultaneous inner conjugation. For a nonabelian simple group, absence of such a marking proves absence of every string C-group marking.*

Proof. Conjugate the first generator to its chosen class representative. Inductively, a conjugation preserving the already fixed ordered prefix is exactly an element of $Z(p)$. String commutation places the next generator in $E(p)$; independence excludes $H(p)$, and connectedness requires its adjacent product order to exceed two. The candidate set is invariant under $Z(p)$, so selecting one representative per orbit loses no possible continuation. The independence and intersection tests reject only prefixes that cannot occur in a C-group. At every recursion the generated subgroup strictly increases. Its order at least doubles, so no branch has unbounded length. This proves termination and exhaustive coverage, without imposing a conjectured rank cutoff. Finally, a commuting adjacent pair in a C-group splits it as a direct product of two nontrivial subgroups, by string commutation and the intersection property. Simplicity excludes this possibility. □

The full intersection test literally constructs $G_I = \langle t_i : i \in I \rangle$ for every subset and checks $G_I \cap G_J = G_{I \cap J}$ for every pair. Thus no maximal-parabolic sufficiency criterion or subgroup-order inference is silently substituted. Empty subgroups are constructed explicitly. At each node the program also checks that the conjugacy orbits cover the whole candidate set disjointly.

3 Exact models and results

Alternating groups use their natural permutation models; the three Mathieu groups use GAP's named standard permutation constructors. The McLaughlin group is the named primitive catalogue group PrimitiveGroup(275, 1). Their identification with these named groups is retained from the standard library. Fresh groups made only from copies of the actual permutation generators have no cached size, stabilizer chain or ambient parent. Verified stabilizer chains on these fresh groups, with no supplied size or upper bound, check their exact orders. The library versions and full input provenance are bound separately; order agreement alone is not asserted to identify an arbitrary permutation group.

For $\mathrm{PSU}_4(3)$ and $\mathrm{PSU}_5(2)$, construct the standard $\mathrm{SU}_n(q)$ generators over $\mathbb{F}_{q^2}$ and obtain their invariant Hermitian form. The program checks its Hermitian symmetry and nonsingularity, each generator's determinant one and isometry equation, and constructs *all* isotropic projective points. A canonical representative has first nonzero coordinate one. There are respectively 280 and 165 points. Fresh uncached copies of their permutation actions have verified orders respectively 3,265,920 and 13,685,760; its kernel has order $\gcd(n, q+1)$. These equal the classical projective-unitary order formulas. Together with the containment established by the isometry equations,

the order equality identifies the full projective unitary image. The standard scalar-center and order formulas are retained classical inputs.

The following table reports the whole prefix counts, by length, in the completed search. These are counts of conjugacy representatives of proper prefixes, not counts of polytopes or abstract subgroup isomorphism classes. No longer prefix exists in these runs; this is an output of exhaustion.

Group	degree	order	proper-prefix counts
A_6	6	360	$1, 6, 18$
A_7	7	2,520	$1, 6, 28, 1$
M_{11}	11	7,920	$1, 6, 32, 3$
M_{22}	22	443,520	$1, 8, 178, 40$
M_{23}	23	10,200,960	$1, 6, 166, 77, 4$
McL	275	898,128,000	$1, 4, 132, 65, 2$
$\mathrm{PSU}_4(3)$	280	3,265,920	$1, 9, 185, 36, 2$
$\mathrm{PSU}_5(2)$	165	13,685,760	$2, 13, 145, 26, 2$

Every row has zero generating C-group markings. The positive A_5 control has proper-prefix counts $1, 3$ and yields six generating rank-three markings up to inner conjugacy. These finite results, the exhaustive algorithm, and its group models are required computational premises of the eight-case exclusion. They are not optional validation of a universal written theorem.

The implementation uses GAP permutation-group computation. Its guaranteed verification option is

`DefaultStabChainOptions.random=1000`.

The marking enumeration is exhaustive; internal stabilizer-chain construction may use random choices. No claim of random-free group-library internals is made. The proof assumes the correctness of the explicitly named GAP operations and group library. No full Lean verification is claimed.

4 The complete principal contract

Theorem 3. *A nonabelian finite simple group admits a string C-group representation if and only if it is not one of the isomorphism types in (1).*

Proof. The retained generation classification and Lemma 1 give a rank-three representation for every group outside (1). For the two projective dimension-three families and both even-characteristic dimension-four families, use the full geometric companion. The eight remaining cases are excluded by Proposition 2 and the complete finite certificate. These alternatives exhaust exactly the source list. □

Theorem 3 is a recovery of the source's principal result, not a new classification or a priority claim. We also recover its projective form theorem with an explicit characteristic-two proof and a direct rank-four form argument. The source's missing finite implementation has been replaced by the whole auditable computation above; a bare assertion that a computer can check the groups is not retained as a theorem. Its positive classification and the classical structural inputs remain attributed and acquired externally. The previous odd-characteristic four-dimensional phase-nine and phase-eighteen notes are contextual prior work, not a proof of these characteristic-two cases; no circular appeal is made to the L024 application, which itself cites L011. There is no new question solution, numerical factor, or complete formalization.

References

[1] D. Leemans and A. Vandenschrick, *On simple groups that admit a string C-group representation*, fixed six-page author manuscript, `https://dipot.ulb.ac.be/dspace/bitstream/2013/397490/3/simpleGroups6.pdf`. Its introduction supplies the exact Nuzhin–Mazurov attribution and the Jones–Mačaj correction; references [5]–[9] identify the retained papers.

[2] M. Conder and D. Oliveros, *The intersection condition for regular polytopes*, J. Combin. Theory A **120** (2013), 1291–1304.

[3] The GAP Group, *GAP*, version 4.15.1, permutation-group manual, especially stabilizer-chain verification, `https://gap-system.github.io/gap/doc/ref/chap43.html`.